

Layers, Depth, and Forward Propagation

Deep Learning

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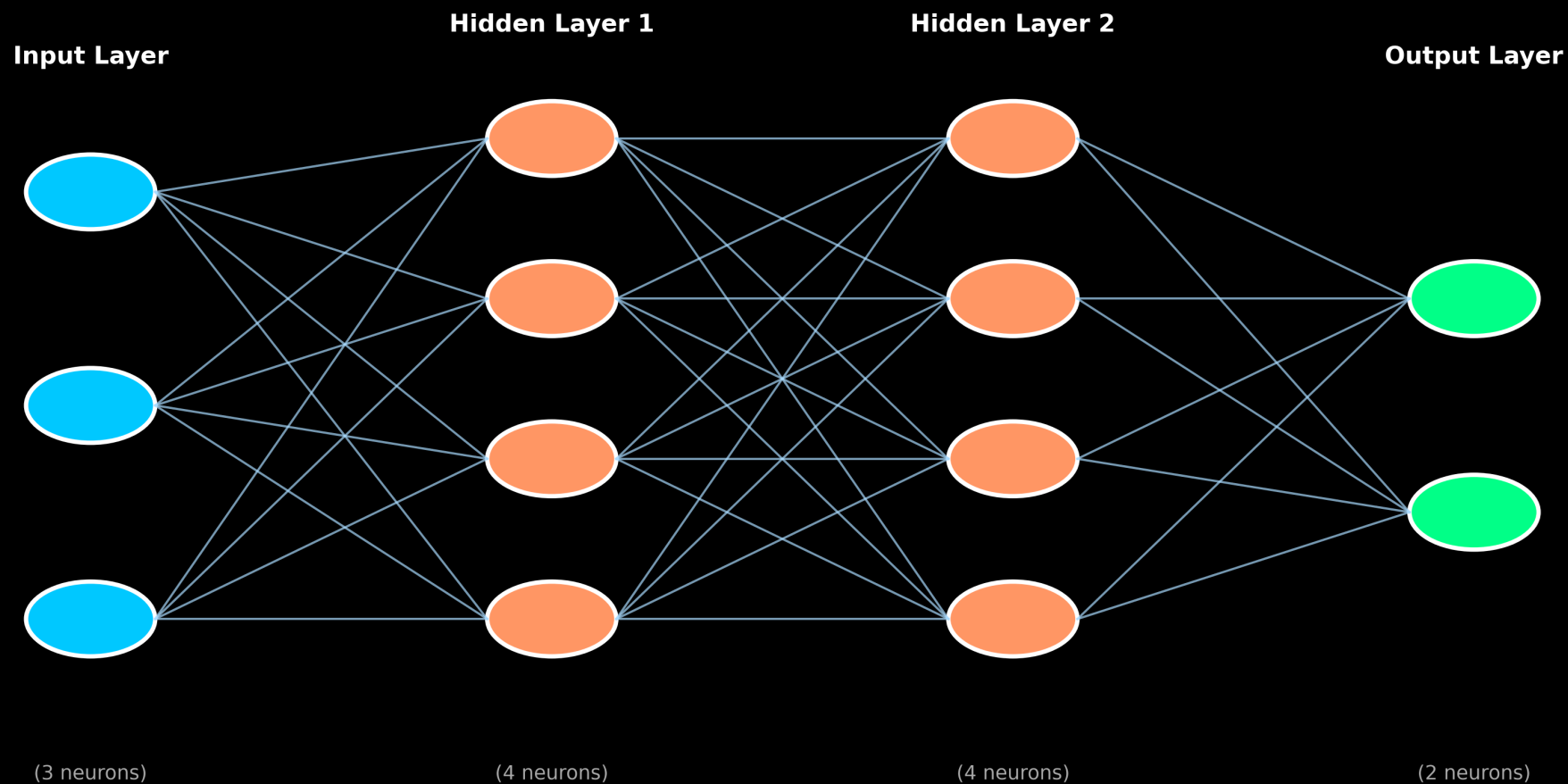
Contents of This Video

In this video, we will cover:

- Input, hidden, and output layers
- Network depth vs. width
- Forward propagation step by step
- A complete numerical example
- Why depth enables learning hierarchical features

Neural Network Architecture

A Neural Network with 2 Hidden Layers



Layer Types

Input Layer:

- Receives the raw features
- No computation, just passes data forward
- Not counted when stating network depth

Hidden Layers:

- Perform the core computations
- Learn intermediate representations
- Each neuron has weights, bias, and activation

Output Layer:

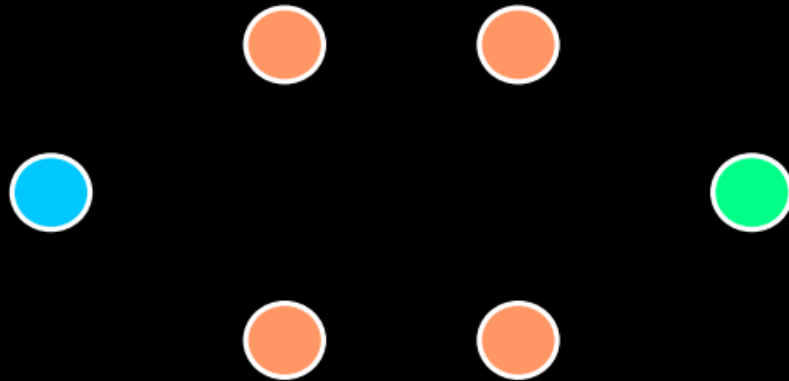
- Produces the final prediction
- Activation depends on task (sigmoid, softmax, linear)

Depth vs. Width

Depth = Number of Layers

- “Deep” learning = many layers
- Enables hierarchical feature learning
- Example: 2-layer network has 2 weight matrices

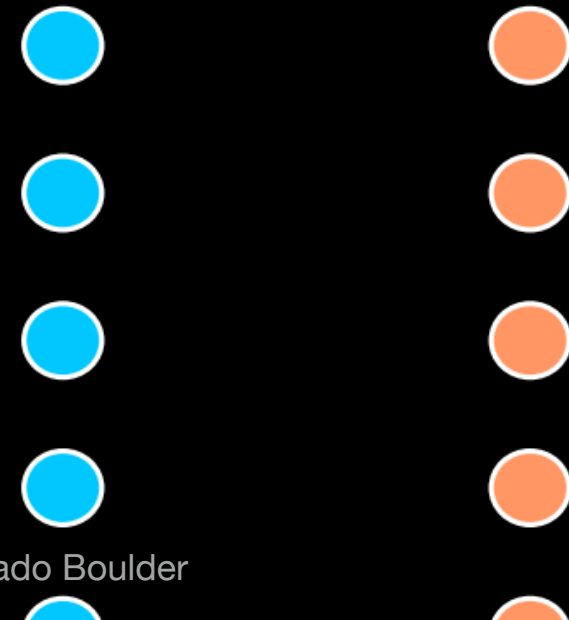
Deep (3 layers)



Width = Neurons per Layer

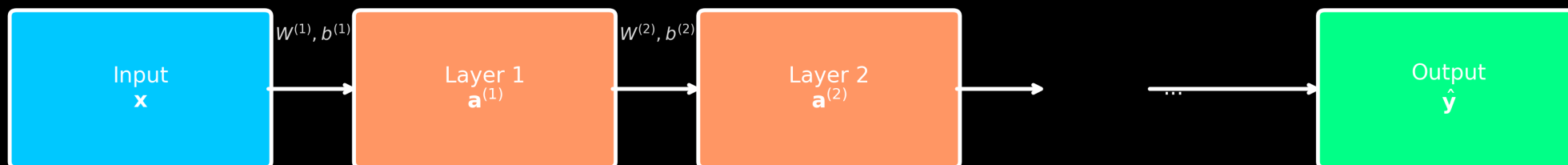
- More neurons = more capacity per layer
- Can be “wide” without being “deep”
- Example: 256 neurons in a hidden layer

Wide (5 neurons/layer)



Forward Propagation: The Big Picture

Forward Propagation: Data Flows Left to Right



At each layer:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}$$

$$\mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)})$$

Forward Propagation: Step by Step

Given input $\mathbf{x}$ and a 2-layer network:

Layer 1:

$$\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$$

$$\mathbf{a}^{(1)} = \text{ReLU}(\mathbf{z}^{(1)})$$

Layer 2 (Output):

$$\mathbf{z}^{(2)} = \mathbf{W}^{(2)}\mathbf{a}^{(1)} + \mathbf{b}^{(2)}$$

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}^{(2)})$$

Numerical Example: Setup

Input: $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (2 features)

Layer 1 (2 neurons):

$$\mathbf{W}^{(1)} = \begin{bmatrix} 0.5 & -0.3 \\ 0.2 & 0.4 \end{bmatrix}, \quad \mathbf{b}^{(1)} = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

Layer 2 (1 neuron, output):

$$\mathbf{W}^{(2)} = \begin{bmatrix} 0.6 & 0.8 \end{bmatrix}, \quad b^{(2)} = 0.2$$

Numerical Example: Layer 1

Step 1: Compute pre-activation

$$\begin{aligned}\mathbf{z}^{(1)} &= \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} = \begin{bmatrix} 0.5 & -0.3 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5(1) + (-0.3)(2) \\ 0.2(1) + 0.4(2) \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 1.0 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.9 \end{bmatrix}\end{aligned}$$

Step 2: Apply ReLU

$$\mathbf{a}^{(1)} = \text{ReLU}(\mathbf{z}^{(1)}) = \text{ReLU} \begin{bmatrix} 0 \\ 0.9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.9 \end{bmatrix}$$

Numerical Example: Layer 2 (Output)

Step 3: Compute pre-activation

$$\begin{aligned} z^{(2)} &= W^{(2)} \mathbf{a}^{(1)} + b^{(2)} = \begin{bmatrix} 0.6 & 0.8 \end{bmatrix} \begin{bmatrix} 0 \\ 0.9 \end{bmatrix} + 0.2 \\ &= 0.6(0) + 0.8(0.9) + 0.2 = 0 + 0.72 + 0.2 = 0.92 \end{aligned}$$

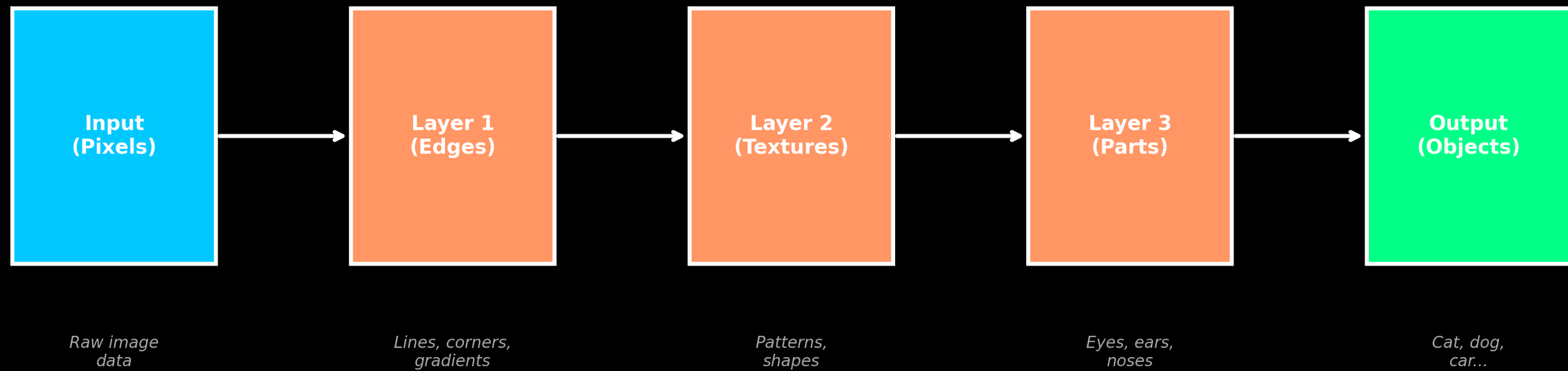
Step 4: Apply sigmoid

$$\hat{y} = \sigma(z^{(2)}) = \frac{1}{1 + e^{-0.92}} \approx 0.715$$

Interpretation: The network predicts ~72% probability of class 1

Why Depth Matters: Hierarchical Features

Hierarchical Feature Learning in Deep Networks



What We've Covered

In this video, we've learned:

- Neural networks have input, hidden, and output layers
- Depth = number of layers; width = neurons per layer
- Forward propagation: compute $z = Wx + b$, then apply activation
- A complete numerical example tracing data through 2 layers
- Deep networks learn hierarchical representations automatically